

# **Direct Numerical Simulations of Mixed-Phase Cloud Glaciation Caused by Turbulent Mixing**

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## **Key Points:**

- Turbulence enhances glaciation if the ice number density is high but has no effect on glaciation if the ice number density is low.
- More but smaller ice crystals promote glaciation compared to fewer but larger ice crystals for the same initial ice mass.
- Hexagonal ice crystals grow preferentially along the shorter dimension and their non-isotropic growth suppresses glaciation compared to spherical ice crystals.

## Abstract

Glaciation of turbulent mixed-phase clouds is examined at the microscale using direct numerical simulations and Lagrangian particle tracking. The effects of flow turbulence intensity, ice number density, and ice crystal aspect ratio on glaciation are investigated. The results show that turbulence enhances glaciation if the ice number density is high. When the ice number density is low, variation of the turbulence intensity does not appear to affect the rate of glaciation. For the same initial ice mass, more but smaller ice crystals promote glaciation compared to fewer but larger ice crystals. It is found that hexagonal ice crystals grow preferentially along the shorter dimension and their non-isotropic growth suppresses glaciation compared to spherical ice crystals. Turbulent mixing influences both the mean and fluctuating thermodynamic properties in our model and mixed-phase cloud glaciation depends on the microscale details of turbulence-cloud-ice crystal interactions.

## 1 Introduction

The saturation vapor pressure with respect to liquid water is higher than that with respect to ice. If the environmental supersaturation level is such that it is supersaturated with respect to ice but subsaturated with respect to water, ice crystals will grow while cloud droplets will shrink. If this continues, a mixed-phase cloud will turn into an ice cloud or glaciates (Korolev et al., 2003). This is the Wegener-Bergeron-Findeisen (WBF) process (Storelvmo & Tan, 2015), also commonly known as glaciation. But mixed-phase clouds have been found to be stable (Morrison et al., 2012), meaning that they do not fully glaciates and remain in a mixed phase condition. To understand the glaciation and persistence of mixed-phase clouds, the interactions between ambient turbulence, cloud droplets, and ice crystals need to be studied at the microscale.

The WBF mechanism was studied analytically by Korolev (2007) who showed that water vapor mass can increase or decrease during this process. An increase of water vapor means that evaporation of cloud droplets is faster than deposition of ice crystals while a decrease of water vapor indicates the opposite. The analytical investigation by Pinksly et al. (2014) suggested that ice crystal number density and initial liquid water mass strongly influence the rate of glaciation. Mixed-phase clouds had been studied numerically. Hill et al. (2013) studied the growth of cloud and ice using a large eddy simulation (LES) model. The authors reported that liquid water mass in a mixed-phase cloud decreases if the ice number density and ice crystal radius increase. Yang et al. (2015) applied LES to examine the growth of ice crystals in a mixed-phase cloud and found that a fraction of ice crystals get trapped in large eddies and grow much bigger than the rest. In their LES investigation, Wang et al. (2024) showed that complete glaciation of a mixed-phase cloud is often restricted by flow turbulence. Chen et al. (2024) studied the evolution of mixed-phase clouds using direct numerical simulations (DNS) and reported that size distributions of cloud droplets broaden more compared to those of ice crystals. They also suggested that initial liquid water mass should be high enough for mixed-phase conditions to persist. Sarnitsky et al. (2024) used DNS to investigate mixed-phase clouds and found that ambient turbulence affects ice crystal growth and thus glaciation.

The geometry of ice crystals is important, particularly at the microscale. Using a theoretical approach, Chen and Lamb (1994) showed that aspect ratio (height/base) of ice crystals changes as ice mass grows with time. In a laboratory investigation, Bailey and Hallett (2003) showed that the complexity of shape for plate-like ice crystals increases if the supersaturation with respect to ice is higher. Sheridan et al. (2009) suggested that growth is more isometric if ice crystals are larger initially but more twisted if they are smaller. Sulia and Harrington (2011) found that glaciation

rate can be under-predicted by nearly an order of magnitude if equivalent spheres are assumed in place of precise ice crystal shape.

In the present study, we implement DNS with Lagrangian tracking of cloud droplets and ice crystals. The temperature and vapor mixing ratio fields are transported by the turbulent velocity field in Eulerian framework. The cloudy (supersaturated) and clear-air (subsaturated) regions are initially separated by a sharp interface (Sharfudin et al., 2025) and subsequent turbulent mixing is simulated. We vary the turbulent kinetic energy (TKE) dissipation rate and number density of ice crystals to assess their impacts on glaciation. The growth profiles of spherical and hexagonal ice crystals are also compared.

## 2 Governing Equations

The turbulent flow of air is governed by the incompressible Navier-Stokes equations with Boussinesq approximation as:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho_a} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{F}_c, \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0. \quad (2)$$

Here,  $\mathbf{u}$  is the instantaneous air velocity vector,  $\nu$  is the kinematic viscosity of air,  $p$  is the instantaneous air pressure and  $\rho_a$  is the air density.  $\mathbf{F}_c$  is an external force to sustain turbulence. The definition of  $\mathbf{F}_c$  can be found in Sharfuddin et al. (2025). Two thermodynamic fields, namely temperature ( $T$ ) and water vapor mixing ratio ( $q_v$ ), are present in our model. Their transport equations are:

$$\frac{\partial T}{\partial t} + (\mathbf{u} \cdot \nabla) T = \frac{L_w}{c_p} C_w + \frac{L_{ice}}{c_p} C_{ice} + \mu_T \nabla^2 T, \quad (3)$$

$$\frac{\partial q_v}{\partial t} + (\mathbf{u} \cdot \nabla) q_v = -(C_w + C_{ice}) + \mu_v \nabla^2 q_v. \quad (4)$$

Above,  $L_w$  and  $L_{ice}$  are specific latent heats of condensation and deposition, respectively.  $C_w$  is the time rate of mass exchange between vapor and liquid while  $C_{ice}$  denotes the same between vapor and ice. Also,  $\mu_T$  and  $\mu_v$  are molecular diffusivities of temperature and vapor, respectively. The cloud droplets and ice crystals are tracked individually in Lagrangian fashion. The equations for the position ( $\mathbf{X}$ ) and velocity ( $\mathbf{V}$ ) vectors can be written as:

$$\frac{d\mathbf{X}_i(t)}{dt} = \mathbf{V}_i(t), \quad \frac{d\mathbf{V}_i(t)}{dt} = \frac{1}{(\tau_p)_i} [\mathbf{u}_i(\mathbf{X}, t) - \mathbf{V}_i(t)] + \mathbf{g}, \quad (\tau_p)_i = \frac{2\rho_i(r_i)^2}{9\rho_a\nu}. \quad (5)$$

where the subscript ' $i$ ' denotes the  $i$ -th particle (cloud droplet or ice crystal).  $\mathbf{u}_i$  is the instantaneous fluid velocity vector obtained at the position of particle  $i$  through a trilinear interpolation of the Eulerian field and  $\tau_p$  is the particle response time. The temporal change in particle radius  $r$  can be written as:

$$\frac{dr_i(t)}{dt} = \frac{G}{r_i(t)} (S_e(\mathbf{X}, t))_i. \quad (6)$$

where  $S_e$  is the environmental supersaturation that depends on the saturation vapor mixing ratio  $q_{v,s}$  and the saturation vapor pressure  $p_{sat}$ .

$$S_e = \frac{q_v}{q_{v,s}} - 1 = \frac{q_v}{621.97 \frac{p_{sat}}{p - p_{sat}}} - 1, \quad (7)$$

$p_{sat}$  can be calculated with respect to water or ice as (Huang, 2018)

$$p_{sat} = \begin{cases} p_{sat,w} = 611.2 \times \exp\left(\frac{17.67(T - 273.15)}{T - 29.65}\right) \\ p_{sat,ice} = 611.2 \times \exp\left(\frac{22.46(T - 273.15)}{T - 0.53}\right) \end{cases} \quad (8)$$

Conveniently, the environmental supersaturation and particle radius can be written as:

$$S_e = \begin{cases} S_{e,w} & \text{if } p_{sat} = p_{sat,w} \\ S_{e,ice} & \text{if } p_{sat} = p_{sat,ice} \end{cases}, \quad r = \begin{cases} r_w & \text{if } S_e = S_{e,w} \\ r_{ice} & \text{if } S_e = S_{e,ice} \end{cases} \quad (9)$$

The growth parameters for cloud droplets and ice crystals in Equation (6) are

$$G_w = \frac{1}{\frac{L_w \rho_w}{k_T T} \left( \frac{L_w}{R_v T} - 1 \right) + \frac{\rho_w R_v T}{\mu_v p_{sat,w}}}, \quad G_{ice} = \frac{1}{\frac{L_{ice} \rho_{ice}}{k_T T} \left( \frac{L_{ice}}{R_v T} - 1 \right) + \frac{\rho_{ice} R_v T}{\mu_v p_{sat,ice}}} \quad (10)$$

With the foregoing,  $C_w$  and  $C_{ice}$  can be determined as

$$C_w(\mathbf{x}, t) = \frac{4\pi \rho_w G_w}{\rho_a a^3} \sum_{i=1}^{n_w} \left( r_w(t) S_{e,w}(\mathbf{X}, t) \right)_i, \quad (11)$$

$$C_{ice}(\mathbf{x}, t) = \frac{4\pi \rho_{ice} G_{ice}}{\rho_a a^3} \sum_{i=1}^{n_{ice}} \left( r_{ice}(t) S_{e,ice}(\mathbf{X}, t) \right)_i. \quad (12)$$

where  $\rho_w$  and  $\rho_{ice}$  are the densities of liquid and ice phases, respectively.  $n_w$  is the number of cloud droplets in a grid cell that has a volume of  $a^3$  and  $n_{ice}$  is the same for ice crystals. Note that  $r_{ice}$  is the equivalent radius of ice crystals if they are not spherical. For a regular hexagon shape, let the aspect ratio be  $h/b$ , where  $h$  is the height and  $b$  is the base. Then the  $r_{ice}$  is found as

$$\frac{4}{3}\pi r_{ice}^3 = \frac{3\sqrt{3}}{2} b^2 h, \quad r_{ice}^3 = 0.62(b^2 h), \quad r_{ice} = 0.8527(b^2 h)^{1/3}. \quad (13)$$

The ice crystal shape is hexagonal column if  $h/b > 1$  and hexagonal plate if  $h/b < 1$ . We see in Equation (6) that particle growth rate  $dr/dt$  is inversely proportional to radius  $r$ . For example, the growth rate of base ( $db/dt$ ) is twice that of height ( $dh/dt$ ) when  $h/b = 2$ . To account for different growth rates along the base and height of hexagonal ice crystals, we solve:

$$\frac{db_i(t)}{dt} = \frac{G_{ice}}{b_i(t)} \left( S_{e,ice}(\mathbf{X}, t) \right)_i, \quad \frac{dh_i(t)}{dt} = \frac{G_{ice}}{h_i(t)} \left( S_{e,ice}(\mathbf{X}, t) \right)_i. \quad (14)$$

Equation (14) gives the updated  $b$  and  $h$ . Then the  $r_{ice}$  is calculated using Equation (13). Note that the  $S_{e,ice}$  is calculated at the center of each ice crystal. It is assumed that any changes in the  $S_{e,ice}$  are negligible at the  $\mu\text{m}$  scale. Moreover, ice crystals grow solely through diffusion in our model and processes like ice nucleation, secondary ice production, and ventilation are neglected.

### 3 Numerical Details

To vary the flow turbulence intensity, we force different wavenumber bands as discussed in Sharfuddin et al. (2025). Specifically, we use the wavenumber bands:  $1 \leq k_f/k_{min} \leq 3.0$  and  $1 \leq k_f/k_{min} \leq 16.18$  to generate ‘low’ and ‘high’ turbulence levels and denote them as ‘L’ and ‘H’, respectively. Here,  $k_f$  is the forced wavenumber and  $k_{min} = 2\pi/L = 12.27$  is the minimum wavenumber using a domain length  $L = 0.512$  m. The turbulent kinetic energy (TKE) dissipation rates ( $\epsilon$ ) for ‘low’ and ‘high’ turbulence levels are  $0.01388 \text{ m}^2/\text{s}^3$  and  $0.00037 \text{ m}^2/\text{s}^3$ , respectively and the Kolmogorov length scales ( $\eta$ ) are  $0.00092$  m and  $0.00226$  m, respectively.

The physical model simulated is that of a cubic domain inside which turbulence, cloud droplets, and ice crystals interact. The domain volume is  $0.512^3 \text{ m}^3$  and the boundary conditions are triply periodic. At the start of simulations, the domain is divided equally into cloudy and clear-air regions. Cloud droplets and ice crystals are randomly placed in the cloudy region only and subsequent turbulent mixing causes them to spread to all over the domain. A slab-like cloud setup is assumed for the initial field to enable a sharp transition in the vapor mixing ratio and temperature at the cloud/clear-air interface (Sharfuddin et al., 2025). The initial vapor mixing ratio field is:

$$q_v(z, t = 0) = \begin{cases} q_v^{max}, & d - L \times 0.5 \times 0.5 \leq z < d + L \times 0.5 \times 0.5 \\ q_{v,e}, & \text{elsewhere} \end{cases} \quad (15)$$

Above,  $q_v^{max} = 2.367 \text{ g/kg}$ ,  $q_{v,e} = 1.90 \text{ g/kg}$ , and  $d \equiv L/2$  is the width of the cloud slab. The temperature field is initialized as:

$$T(t = 0) = \langle T(t = 0) \rangle - 0.608 \langle T(t = 0) \rangle (q_v(z, t = 0) - \langle q_v(t = 0) \rangle). \quad (16)$$

We simulate six cases: L-H, H-H, L-L, H-L, L-H-C, and L-H-P, as shown in Table I. The same table also lists the values of relevant parameters used in the simulations. The identifiers ‘L’ and ‘H’ at first position indicate ‘low’ and ‘high’ turbulence intensities. The same identifiers at second position correspond to ‘low’ and ‘high’ ice number densities, respectively. Note that a higher ice number density corresponds to smaller mean ice crystal radius and the vice versa in our model because the initial ice masses are equal for all cases. Also, ice crystals are spherical for the first four cases. The identifiers ‘C’ and ‘P’ at third position signify hexagonal columns and plates, respectively. The number density of cloud droplets is  $59.6 \text{ cm}^{-3}$  for all cases and is much higher than those of ice crystals. This is a realistic scenario for mixed-phase clouds. The number of grid points used is  $256^3$  for all cases. Particles (cloud droplets or ice crystals) are inserted in the model domain after 4 s when turbulence becomes statistically stationary.

Note that our initial setup differs from that of Chen et al. (2024) in the sense that there are two separate regions marked by distinct supersaturation levels. This allows us to evaluate the impact of turbulent mixing in a more realistic setting. Also, our DNS model uses a fifth order finite difference scheme to solve the convective terms in Equations (1), (3), and (4) as sharp interfaces are present (Equations (15) and (16)). Whereas Chen et al. (2024) used a pseudospectral method. More details about our DNS model can be found in Sharfuddin et al. (2025).

**Table 1**

*Details of the test cases*

| Case  | TKE<br>dissipation<br>rate ( $\text{m}^2/\text{s}^3$ ) | Ice number<br>intensity<br>( $\text{cm}^{-3}$ ) | Initial ice<br>crystal radius<br>( $\mu\text{m}$ ) | Initial cloud<br>droplet<br>radius ( $\mu\text{m}$ ) | Aspect ratio<br>( $h/b$ ) of ice<br>crystals |
|-------|--------------------------------------------------------|-------------------------------------------------|----------------------------------------------------|------------------------------------------------------|----------------------------------------------|
| L-H   | 0.01388                                                | 0.011                                           | 7.5                                                | 7.5                                                  | 1.0                                          |
| H-H   | 0.00037                                                | 0.011                                           | 7.5                                                | 7.5                                                  | 1.0                                          |
| L-L   | 0.01388                                                | 0.0011                                          | 16.16                                              | 7.5                                                  | 1.0                                          |
| H-L   | 0.00037                                                | 0.0011                                          | 16.16                                              | 7.5                                                  | 1.0                                          |
| L-H-C | 0.01388                                                | 0.011                                           | 7.5                                                | 7.5                                                  | 2.5                                          |
| L-H-P | 0.01388                                                | 0.011                                           | 7.5                                                | 7.5                                                  | 0.4                                          |

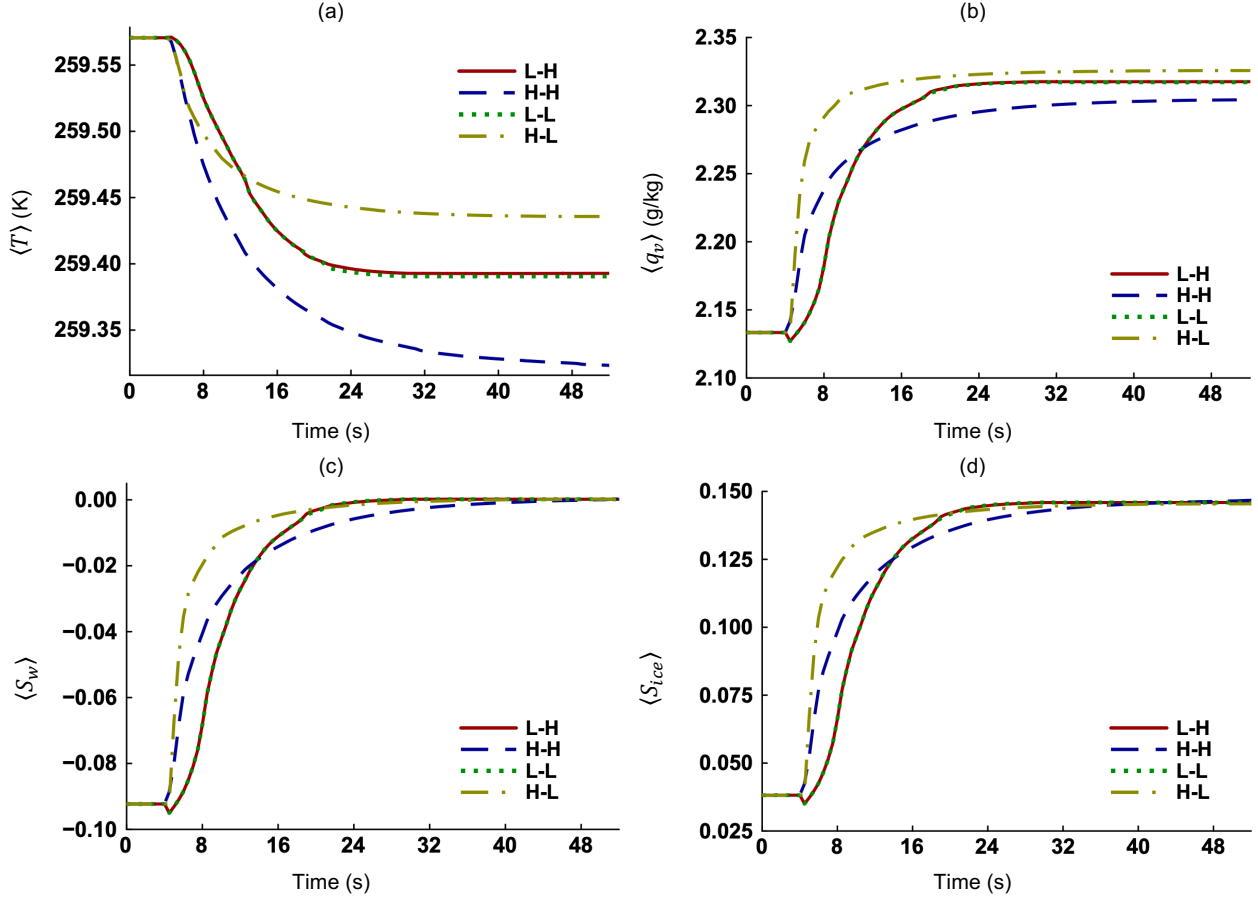
*Parameters and their values*

| Symbol    | Value                 | Unit              | Symbol                     | Value                | Unit              |
|-----------|-----------------------|-------------------|----------------------------|----------------------|-------------------|
| $L_w$     | $2.50 \times 10^6$    | J/kg              | $c_p$                      | 1005.0               | J/kg/K            |
| $L_{ice}$ | $2.83 \times 10^6$    | J/kg              | $\rho_a$                   | 0.7674               | kg/m <sup>3</sup> |
| $\rho_w$  | 1000                  | kg/m <sup>3</sup> | $\rho_{ice}$               | 917                  | kg/m <sup>3</sup> |
| $p$       | 57160                 | N/m <sup>2</sup>  | $\langle T(t = 0) \rangle$ | 259.53               | K                 |
| $R_v$     | 461.5                 | J/kg/K            | $k_T$                      | 0.0238               | W/m/K             |
| $\mu$     | $1.64 \times 10^{-5}$ | m <sup>2</sup> /s | $\mu_T = \mu_v$            | $2.2 \times 10^{-5}$ | m <sup>2</sup> /s |

#### 4 Results and Discussion

The present model is initialized in a way that mean supersaturations with respect to water and ice are negative and positive, respectively. So, ice crystals grow but cloud droplets evaporate, and glaciation takes place. We vary the flow turbulence level, ice number density, and ice crystal aspect ratio. These are represented by the six cases (Table I). The effects on thermodynamics and microphysics are evaluated to understand the glaciation of mixed-phase cloud.

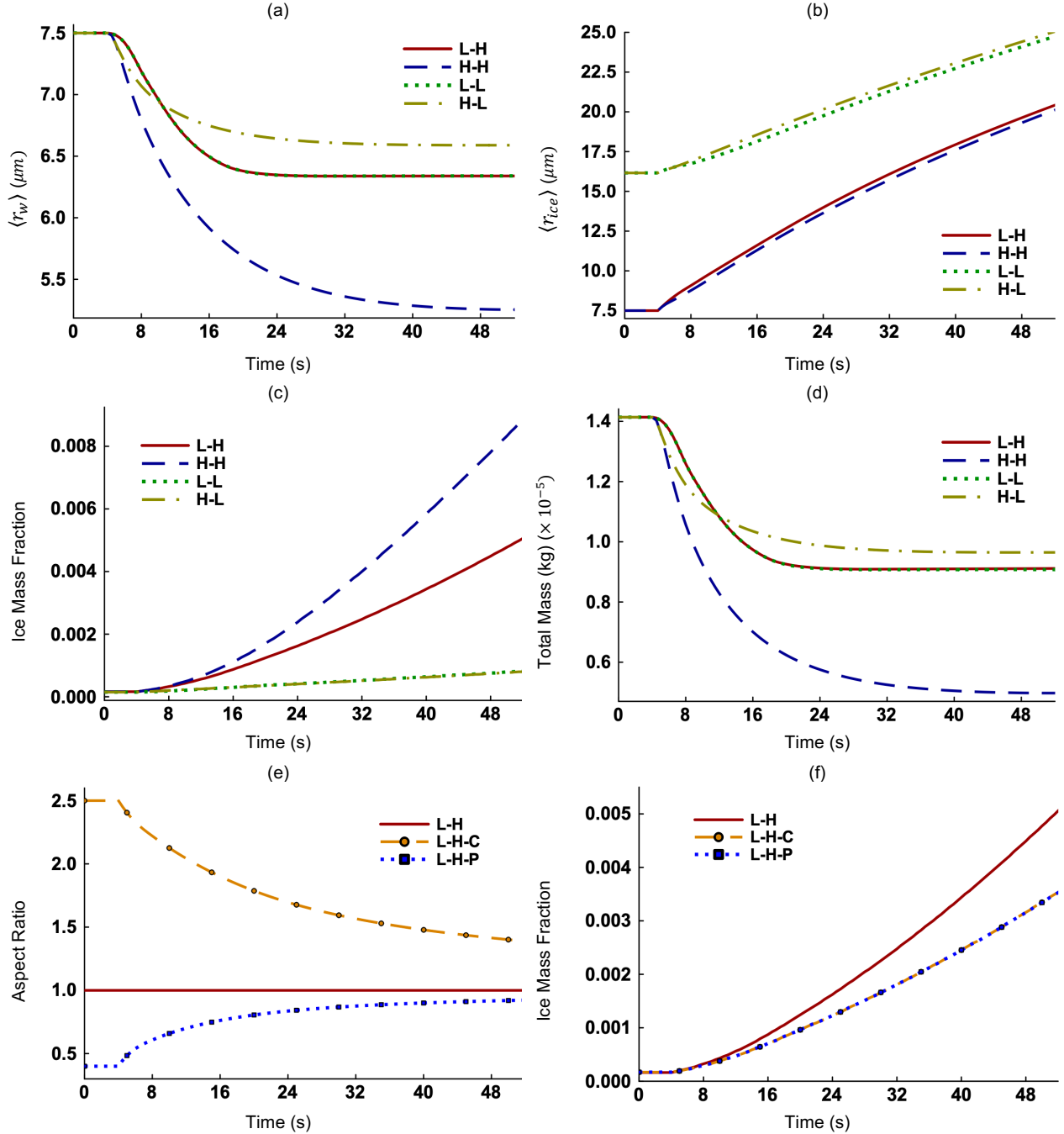
Figures 1(a)-(d) show the temporal evolutions of mean thermodynamic fields. We see that the mean temperature  $\langle T \rangle$  decreases but the mean vapor mixing ratio  $\langle q_v \rangle$  increases with time. Liquid water converts into water vapor through evaporation of cloud droplets while ice crystals grow by absorbing water vapor. So, water vapor increases and decreases at the same time. There is a net increase of water vapor in our simulations (Figure 1(b)) because the mass of cloud droplets far exceeds that of ice crystals. The  $\langle T \rangle$  decreases when the  $\langle q_v \rangle$  increases as latent heat terms have opposite signs in Equations (3) and (4). The mean supersaturations with respect to water  $\langle S_{e,w} \rangle$  and ice  $\langle S_{e,ice} \rangle$  increase with time (Figures 1(c)-(d)) since they are directly proportional to the  $\langle q_v \rangle$ . The mean profiles of all thermodynamic variables tend to asymptote when the  $\langle S_{e,w} \rangle$  approaches zero (saturation) which indicates that the system has reached equilibrium. The increase of  $\langle S_{e,w} \rangle$  from negative to zero is fastest for Case H-L but slowest for Case H-H (Figure 1(c)). The longer the subsaturation (negative  $S_{e,w}$ ) persists, the stronger will be the evaporation. Thus, more ice but smaller crystals enhance evaporation in our model if the turbulence level is high (Case H-H). However, the  $\langle S_{e,w} \rangle$  profiles are identical for Cases L-H and L-L, suggesting that ice number density (or ice crystal size) does not influence evaporation if the turbulence level is low. We also see that the  $\langle S_{e,w} \rangle$  increases sharply for high turbulence cases (H-L and H-H) compared to low turbulence cases (L-L and L-H) at very early times. But the  $\langle S_{e,w} \rangle$  and  $\langle S_{e,ice} \rangle$  profiles become more gradual for Case H-H as simulation progresses, apparently caused by the turbulent transport of more but smaller ice crystals and associated phase change.



**Figure 1.** Time evolution of the (a) mean temperature, (b) mean water vapor mixing ratio, (c) mean environmental supersaturation with respect to water and (d) mean environmental supersaturation with respect to ice.

The time evolutions of the mean radii of cloud droplets ( $\langle r_w \rangle$ ) and ice crystals ( $\langle r_{ice} \rangle$ ) are reported in Figures 2(a)-(b), respectively. Consistent with Figure 1(c), stronger evaporation leads to maximum decrease of the  $\langle r_w \rangle$  for Case H-H (Figure 2(a)) and weaker evaporation corresponds to minimum decrease of the  $\langle r_w \rangle$  for Case H-L. The  $\langle r_{ice} \rangle$  increases almost linearly for all cases since the environment is highly supersaturated with respect to ice (Figure 1(d)). This observation agrees with Chen et al. (2024). Also, the  $\langle r_{ice} \rangle$  increases slightly faster for Case L-H compared to Case H-H, indicating that high turbulence has a retarding effect on ice crystal growth if the ice number density is high. The opposite is true if the ice number density is low (Cases L-L vs H-L). Figure 2(c) depicts that the ice mass fractions are identical initially and they keep increasing as expected during glaciation. The increase is maximum for Case H-H and minimum for Cases L-L and H-L while that for Case L-H falls in between. The ice mass fraction is the ratio of ice mass to total mass of liquid and ice. It increases due to either ice phase growth or evaporative decay of liquid phase. Although ice crystal growth is slightly higher for Case L-H compared to Case H-H (Figure 2(b)), significant evaporation for Case H-H (Figure 2(a)) results in higher ice mass fraction for this case (Figure 2(c)). We see that ice mass fraction increases more slowly for Cases L-L and H-L than the other two cases. Hence, fewer but larger ice crystals correspond to slower glaciation rate while more but smaller ice crystals suggest the opposite. The profiles of total mass (Figure

185 2(d)) resemble those of mean cloud droplet radii (Figure 2(a)) because evaporation is dominant  
 186 over deposition in our model.

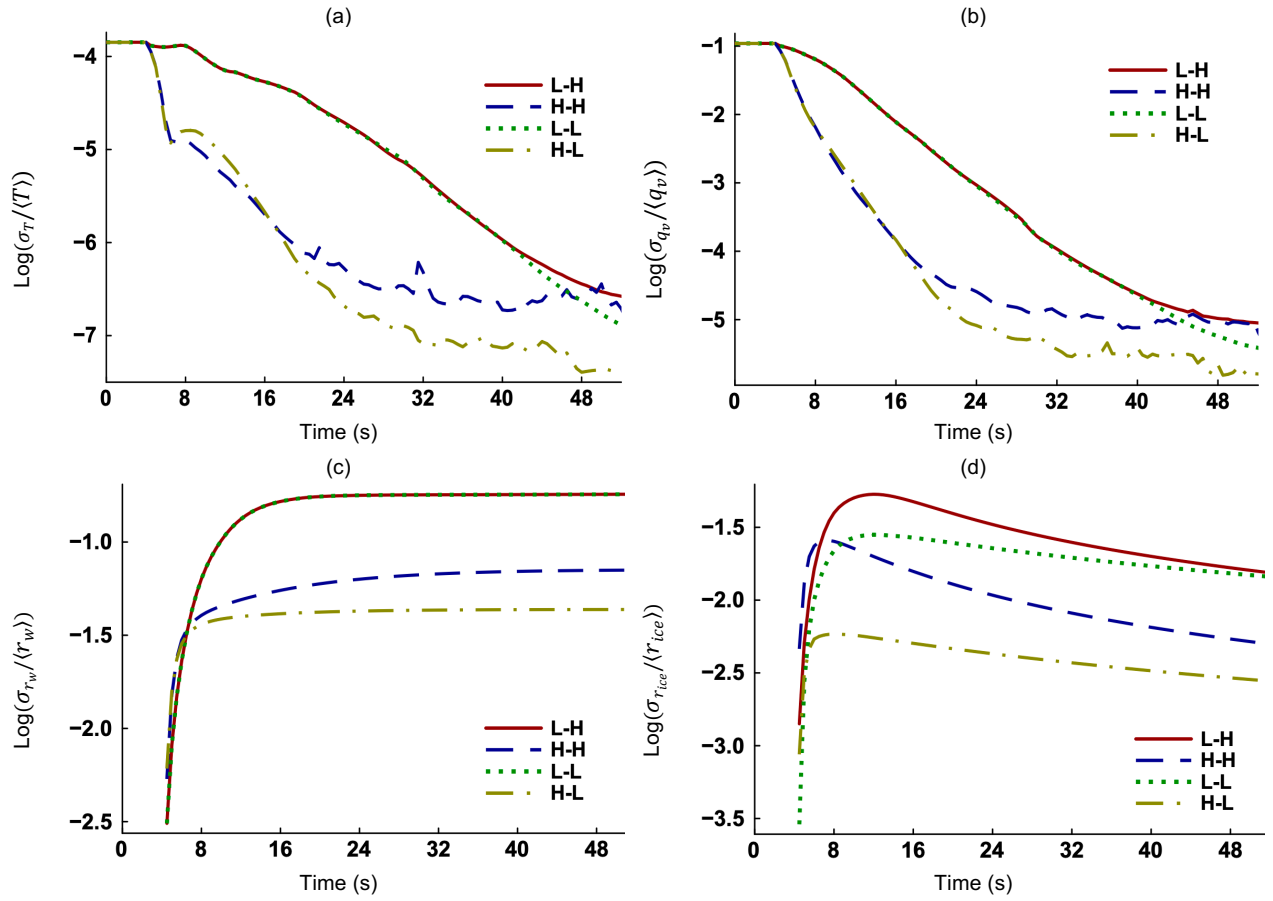


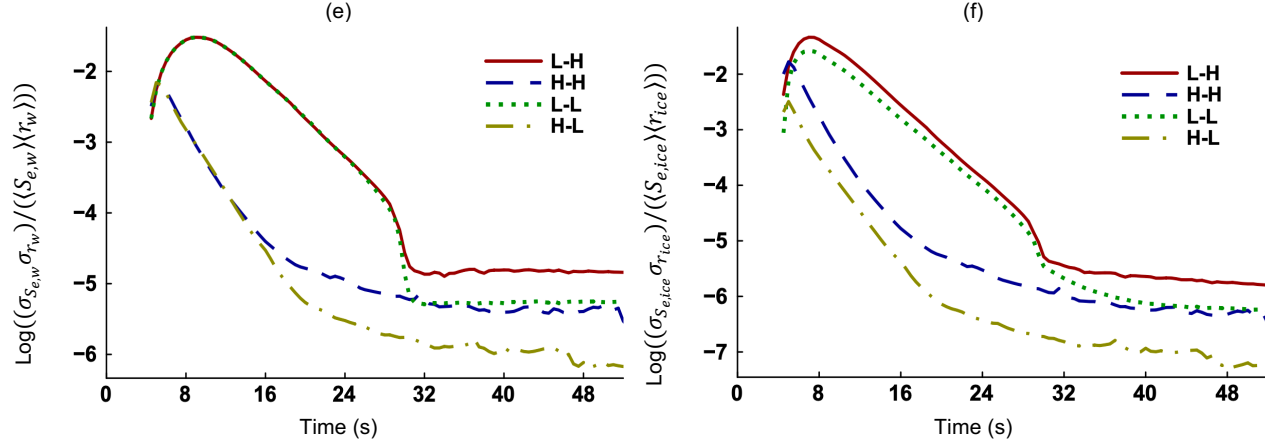
187 **Figure 2.** Time evolution of the (a) mean radius of cloud droplets, (b) mean radius of ice  
 188 crystals, (c) ice mass fraction and (d) total mass of particles, for Cases L-H, H-H, L-L, and H-L.  
 189 Time evolution of the (e) ice crystal aspect ratio and (f) ice mass fraction, for Cases L-H, L-H-C,  
 190 and L-H-P.

191 Figures 2(e)-(f) show the time evolutions of ice crystal aspect ratio and ice mass fraction,  
 192 respectively, for three cases: L-H, L-H-C, and L-H-P. The initial aspect ratios are 1.0, 2.5, and 0.4  
 193 for ice crystals resembling sphere, hexagonal column, and hexagonal plate, respectively. We find



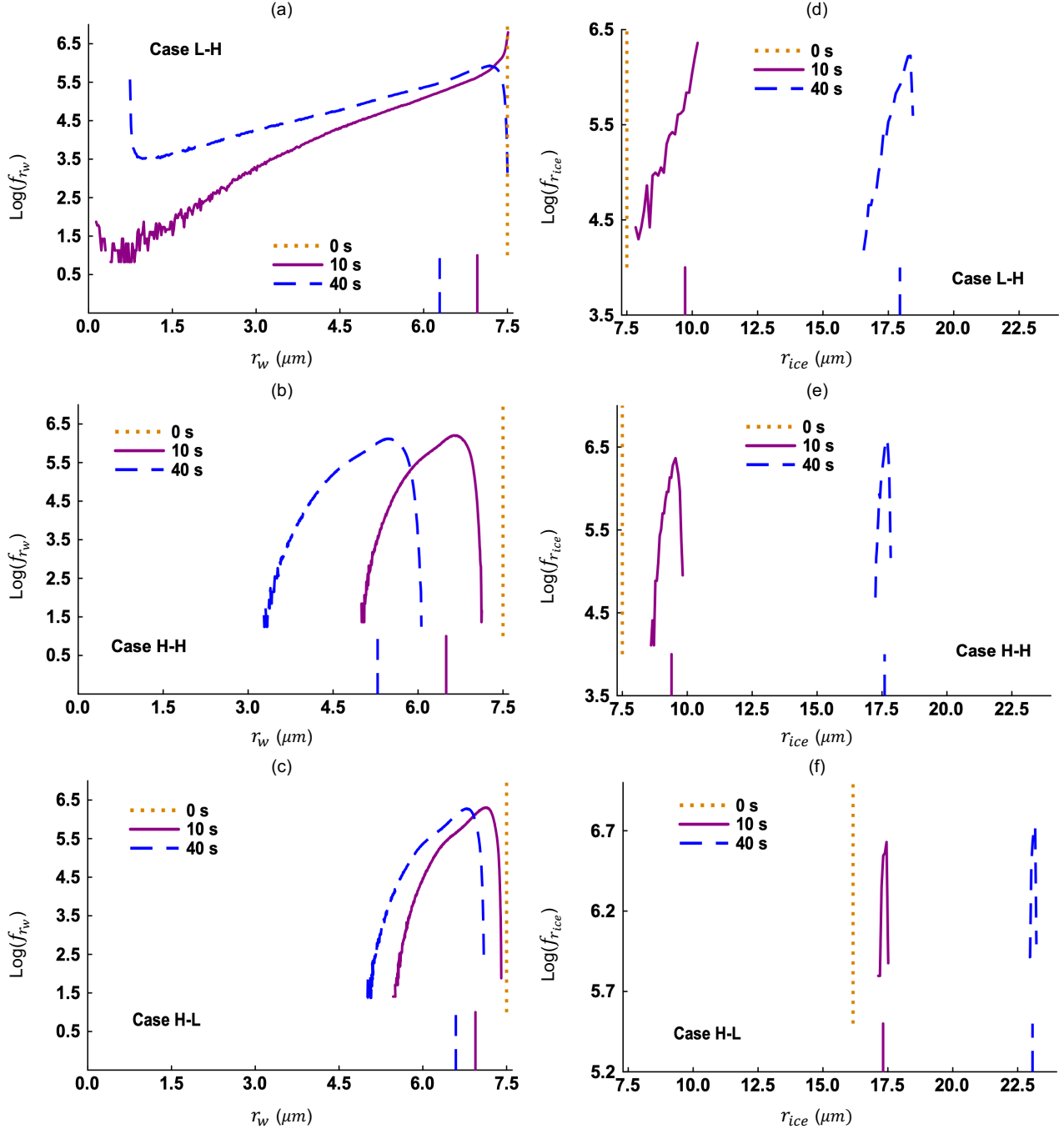
that the aspect ratios of hexagonal column and hexagonal plate converge to unity as simulations progress. This suggests that ice crystal growth is faster along the shorter dimension. The tendency of ice crystals to grow in a way that minimizes surface area-to-volume ratio makes hexagonal ice crystals more spherical with time. As ice number density is much lower than that of cloud droplets in our model, ice crystals do not face competition for water vapor. Locally consumed water vapor, as a result of ice crystal growth, is quickly replenished through evaporation and turbulence. Hence, ice crystals can grow preferentially along the shorter dimension. In real mixed-phase clouds, ice crystals with complex shape exist depending on the temperature, supersaturation, and growth mechanism (Bailey and Hallett, 2003). But under simulated conditions and only diffusional growth assumption, ice crystals evolve to become more regular in our model. Note that the aspect ratio for spherical ice crystals remains unchanged (unity). Figure 2(f) shows that ice mass fraction increases faster when ice crystals are spherical (Case L-H) because isotropic growth of spherical ice crystals is more efficient than anisotropic growth of hexagonal plates and columns.





**Figure 3.** Time evolution of the relative dispersions of (a) temperature ( $\sigma_T/\langle T \rangle$ ), (b) vapor mixing ratio ( $\sigma_{q_v}/\langle q_v \rangle$ ), (c) cloud droplet radii ( $\sigma_{r_w}/\langle r_w \rangle$ ), (d) ice crystal radii ( $\sigma_{r_{ice}}/\langle r_{ice} \rangle$ ). Time evolution of the joint relative dispersions of (e) supersaturation with respect to water and cloud droplet radii ( $((\sigma_{S_{e,w}} \sigma_{r_w}) / ((S_{e,w}) \langle r_w \rangle))$ ), and (f) supersaturation with respect to water ice and ice crystal radii ( $((\sigma_{S_{e,ice}} \sigma_{r_{ice}}) / ((S_{e,ice}) \langle r_{ice} \rangle))$ ). The ordinates are in log scale.

The time evolutions of relative dispersion, which is standard deviation divided by mean, are plotted in Figures 3(a)-(b) for temperature and vapor mixing ratio, respectively. The magnitude of thermodynamic fluctuations is largest at the start because the cloudy and clear-air regions have distinct  $T$  and  $q_v$  values, as specified initially. Both  $\sigma_T/\langle T \rangle$  and  $\sigma_{q_v}/\langle q_v \rangle$  decrease afterwards since turbulent mixing homogenizes thermodynamic fluctuations (Sharfuddin et al., 2025). The higher the turbulence intensity, the quicker the homogenization (Cases H-L and H-H). Figures 3(c)-(d) are analogous to Figures 3(a)-(b) but for cloud droplet and ice crystal radii, respectively. At first, monodisperse cloud droplets and ice crystals are exposed to turbulent fluctuations and their relative dispersions increase. At late times, turbulent fluctuations decrease, ice crystals grow at a relatively uniform rate and the  $\sigma_{r_{ice}}/\langle r_{ice} \rangle$  decreases from its peak for all cases (Figure 3(d)). The  $\sigma_{r_w}/\langle r_w \rangle$  becomes steady instead of decreasing because cloud droplet evaporation is steady at late times (Figure 2(a)) unlike ice crystal growth which continues (Figure 2(b)). We also find that higher turbulence intensity has a retarding effect on the relative dispersions of both cloud droplets and ice crystals as a result of enhanced mixing. The ice number density has an expediting effect on the  $\sigma_{r_{ice}}/\langle r_{ice} \rangle$  for both turbulence levels and on the  $\sigma_{r_w}/\langle r_w \rangle$  if the turbulence level is high (Cases H-H vs H-L). However, it has no effect on the  $\sigma_{r_w}/\langle r_w \rangle$  when the turbulence level is low. The plots of joint relative dispersions (Figures 3(e)-(f)) show that the ice number density starts to influence the  $((\sigma_{S_{e,w}} \sigma_{r_w}) / ((S_{e,w}) \langle r_w \rangle))$  only after the latter has decreased by orders. On the other hand, the  $((\sigma_{S_{e,ice}} \sigma_{r_{ice}}) / ((S_{e,ice}) \langle r_{ice} \rangle))$  differs from the start because the  $\langle r_{ice} \rangle$  varies between Cases L-H and L-L, and between Cases H-H and H-L.



**Figure 4.** Log PDFs of cloud droplet size for Cases (a) L-H, (b) H-H, and (c) H-L. Log PDFs of ice crystal size for Cases (d) L-H, (e) H-H, and (f) H-L. The vertical line segments at the bottom indicate mean values in the respective distributions.

The PDFs of cloud droplet radii are shown in Figures 4(a)-(c) and those of ice crystal radii are shown in Figures 4(d)-(f). The radius PDFs are broader at 10 s compared to 0 s for all cases due to turbulent fluctuations. Between Cases L-H (Figure 4(a)) and H-H (Figure 4(b)), the radius PDFs of cloud droplets are broader for the former at 10 s and 40 s because of less homogenization. Both the cloud and ice PDFs are strongly skewed to the left for Case L-H at 10 s and 40 s. The corresponding cloud PDFs for Case H-H are also negatively skewed, but to a much lesser extent.

The ice PDFs are significantly narrower than the cloud PDFs for all cases because cloud number density is much higher than that of ice crystals in our model. The cloud PDF is narrower, and the mean cloud droplet radius decreases less by 40 s for Case H-L (Figure 4(c)) compared to Case H-H (Figure 4(b)). Hence, evaporation is suppressed along with less variation in cloud droplet radii if ice crystals are fewer in number but larger in size.

## 5 Conclusions

We have examined the glaciation of mixed-phase cloud using a three-dimensional DNS model and a Lagrangian microphysics scheme. We show that flow turbulence influences the mean and fluctuating thermodynamic properties, microphysics of cloud droplets and ice crystals, and thus glaciation. Moreover, the number density, size, and geometry of ice crystals affect glaciation depending on the simulation details. A microscale understanding of the glaciation process has the potential to contribute to reducing uncertainty in the estimation of mixed-phase cloud albedo which depends on the available liquid and ice mass. Again, precise calculation of the cloud's reflectivity (albedo) is required for accurate prediction of Earth's radiation budget. The findings of the present work will be useful in this regard.

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## Open Research

The data that support the findings of this work are available from the corresponding author upon reasonable request.

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